

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 5. Conditional Statements

Lecture: 41. Challenge - Exam Result

Section 1: Lecture Summary

This lecture covers solving two programming challenges using compound conditional statements in Python. The first challenge is to determine whether a single lowercase English letter input is a vowel or a consonant based on a compound condition using logical OR statements. The second challenge is to evaluate exam results from three subject marks (Maths, Physics, Chemistry), each with a pass mark of 45, using compound conditional statements with logical AND to decide pass or fail. The instructor explains the logic behind framing these compound conditions and demonstrates complete Python programs for both challenges, encouraging students to try coding them independently using any Python environment like Jupyter Notebook, PyCharm, or VS Code.

Section 2: Key Concepts and Explanations

- Compound Conditional Statements:

- Compound conditions combine multiple simple conditions using logical operators such as ``or`` and ``and``.
- A condition joined by ``or`` is true if any one of the sub-conditions is true.
- A condition joined by ``and`` is true only if all sub-conditions are true.

- Vowel or Consonant Challenge:

- Vowels in English lowercase alphabets: ``a`, 'e', 'i', 'o', 'u'``.
- To check if a letter is a vowel, use a compound condition:

vowel

consonant

```
if x == 'a' or x == 'e' or x == 'i' or x == 'o' or x == 'u':  
    else:
```

- Else, the letter is considered a consonant.

- Exam Result Challenge:

- Subjects: Maths, Physics, Chemistry.
- Full marks: 100 each.
- Pass marks per subject: 45.
- The student passes only if the marks in all three subjects are greater than or equal to 45.
- Compound condition with `and` ensures all subjects meet the pass criteria:

pass

fail

```
if maths >= 45 and physics >= 45 and chemistry >= 45:  
    else:
```

- Failing in any one subject results in overall fail.

- Input Handling:

- Use the `input()` function to take user input.
- Input values from `input()` are strings by default.
- For numerical inputs (marks), convert input strings to integers using `int()`.

Section 3: Example Code and Use Cases

1. Vowel or Consonant Program

Program to check if a lowercase letter is vowel or consonant

```
x = input('Enter a lowercase letter: ') # Take input as string

if x == 'a' or x == 'e' or x == 'i' or x == 'o' or x == 'u':
    print('Vowel')
else:
    print('Consonant')
```

Explanation: This program takes a single lowercase letter input and checks it against vowels using multiple `or` conditions. If the input matches any vowel, it prints "Vowel," else "Consonant."

2. Exam Result Program

Program to determine exam pass or fail based on marks in three subjects

```
maths = int(input('Enter marks in Maths: '))
physics = int(input('Enter marks in Physics: '))
chemistry = int(input('Enter marks in Chemistry: '))

if maths >= 45 and physics >= 45 and chemistry >= 45:
    print('Pass')
else:
    print('Fail')
```

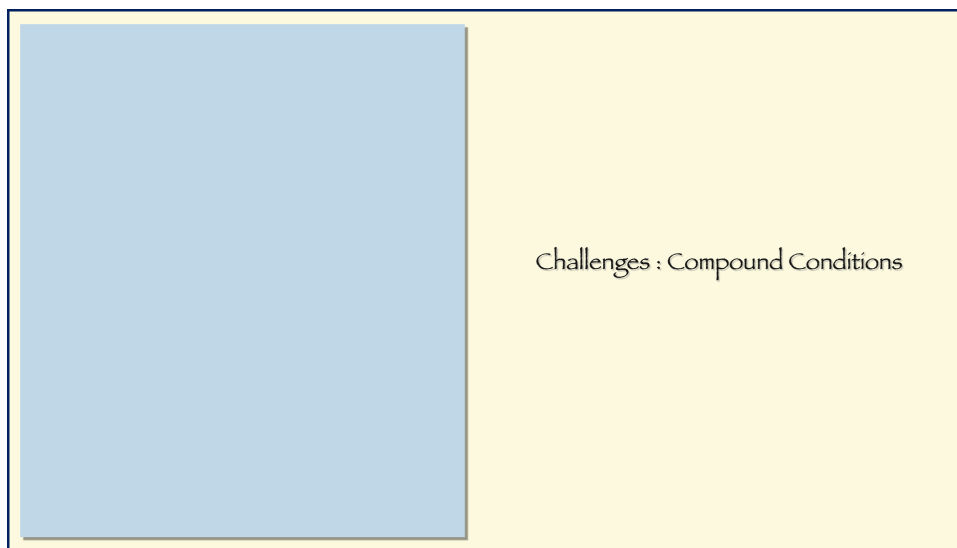
Explanation: This program takes integer marks inputs in three subjects, checks if all are above or equal to 45 using `and` operator. If yes, prints "Pass"; otherwise, "Fail."

Section 4: Key Takeaways

- Compound conditional statements use logical operators ``or`` and ``and`` to combine multiple conditions.
- Use ``or`` to check if any one condition is true.
- Use ``and`` to check if all conditions are true.
- Vowels are the letters ``a', 'e', 'i', 'o', 'u'``; all others in the alphabet are consonants.
- To pass an exam with multiple subjects, each subject's marks must meet or exceed the passing criteria.
- The ``input()`` function returns strings; convert to integer with ``int()`` when working with numeric data.
- Writing clear and complete conditional checks allows building simple decision-making programs.
- Practicing such challenges in environments like Jupyter Notebook or IDEs improves coding skills and understanding of conditionals.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.

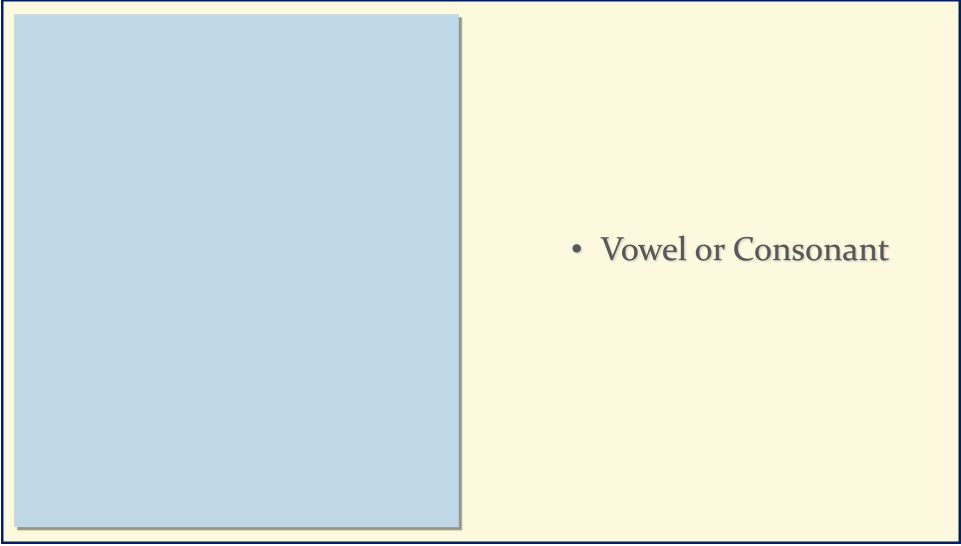


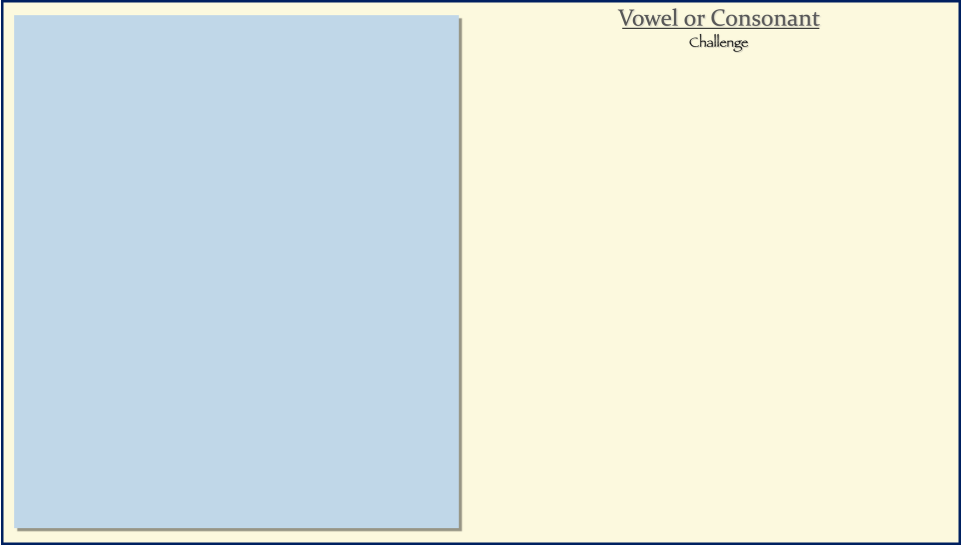
Challenges : Compound Conditions

- Vowel or Consonant

Challenges : Compound Conditions

- Vowel or Consonant
- Exam Result

- 
- Vowel or Consonant



Vowel or Consonant
Challenge

Vowel or Consonant

Challenge

English Alphabets

Vowel or Consonant

Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowel or Consonant

Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowel or Consonant

Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
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Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
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Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
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Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
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v	w	x	y	z		

Vowel or Consonant
Challenge

English Alphabets						
a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

Vowel or Consonant
Challenge

English Alphabets						
a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = 'a'

Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = 'a'

'Vowel'

Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = 'b'

'Consonant'

Vowel or Consonant

Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = ?

Vowel or Consonant

Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = ?

Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = ?

x = ?

Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = ?

x = ?

if x == 'a' :

Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = ?

x = ?

if x=='a' or x=='e':

Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = ?

x = ?

if x=='a' or x=='e' or x=='i':

Vowel or Consonant
Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = ?

x = ?

if x=='a' **or** x=='e' **or** x=='i' **or** x=='o' **or** x=='u' :

Vowel or Consonant
Challenge

English Alphabets							
a	b	c	d	e	f	g	
h	i	j	k	l	m	n	
o	p	q	r	s	t	u	
v	w	x	y	z			

Vowels = {a, e, i, o, u}

letter = ?

x = ?

```
if x=='a' or x=='e' or x=='i' or x=='o' or x=='u':  
    print('Vowel')
```

Vowel or Consonant
Challenge

English Alphabets							
a	b	c	d	e	f	g	
h	i	j	k	l	m	n	
o	p	q	r	s	t	u	
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Vowels = {a, e, i, o, u}

letter = ?

x = ?

```
if x=='a' or x=='e' or x=='i' or x=='o' or x=='u':  
    print('Vowel')  
else:
```

Vowel or Consonant

Challenge

English Alphabets

a	b	c	d	e	f	g
h	i	j	k	l	m	n
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Vowels = {a, e, i, o, u}

letter = ?

x = ?

```
if x=='a' or x=='e' or x=='i' or x=='o' or x=='u':  
    print('Vowel')  
else:  
    print('Consonant')
```

Vowel or Consonant

Challenge

English Alphabets

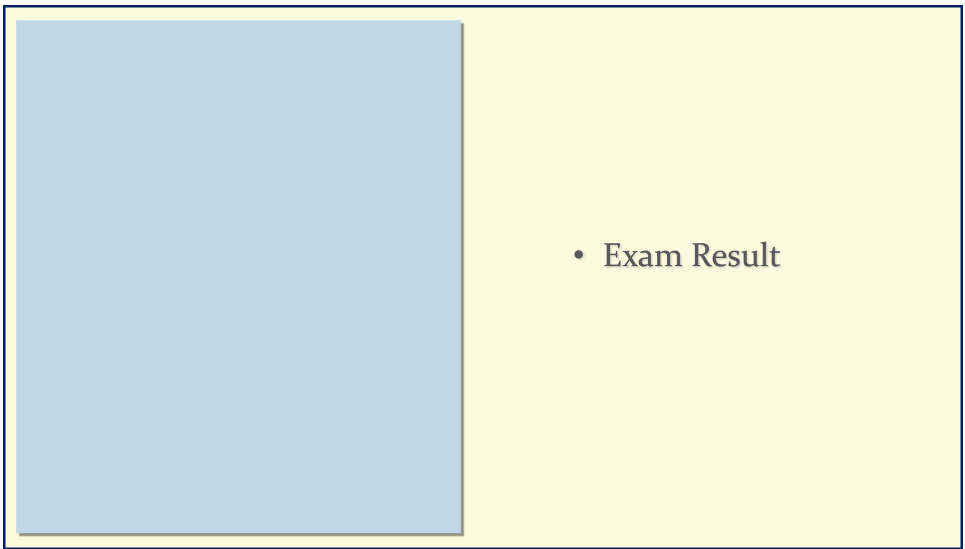
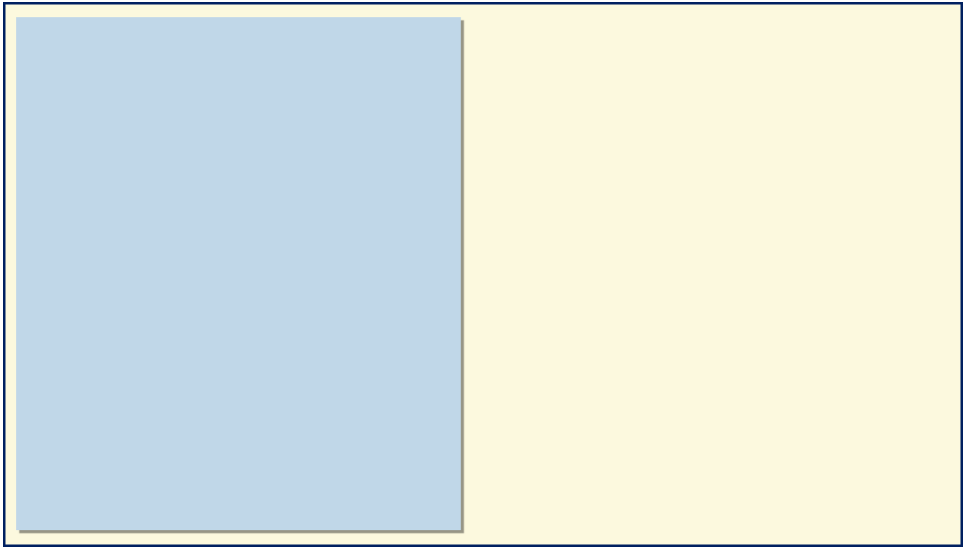
a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	w	x	y	z		

Vowels = {a, e, i, o, u}

letter = ?

x = input('Enter a Lowercase Letter')

```
if x=='a' or x=='e' or x=='i' or x=='o' or x=='u':  
    print('Vowel')  
else:  
    print('Consonant')
```



Exam Result

Challenge



maths

Exam Result

Challenge

Exam Result

Challenge

maths

physics



Exam Result

Challenge

Subject	Score
maths	85%
physics	78%
chem	92%

Challenge



chem

Challenge



physics



Exam Result

Challenge

Subject	Score
maths	85
physics	72
chem	91

Max Marks =100
Pass Marks =45



physics



Exam Result

Challenge

Subject	Max Marks = 100	Pass Marks = 45	Scored
maths	100	45	50
physics	100	45	40
chem	100	45	40



physics



maths = 50

Challenge



physics



maths = 50
physics = 80

Challenge



physics



maths = 50
physics = 80
chem = 65

Exam Result

Challenge



maths



physics



chem

Max Marks =100

Pass Marks =45

maths = 50

physics = 80

chem = 65

print('Pass')

Exam Result

Challenge



maths



physics



chem

Max Marks =100

Pass Marks =45

maths >= 45

Exam Result

Challenge



maths



physics



chem

Max Marks =100

Pass Marks =45

maths >= 45

physics >= 45

Exam Result

Challenge



maths



physics



chem

Max Marks =100

Pass Marks =45

maths >= 45

physics >= 45

chem >= 45

Exam Result

Challenge


maths


physics


chem

Max Marks =100

Pass Marks =45

maths >= 45 and physics >= 45 and chem >= 45

Exam Result

Challenge


maths


physics


chem

Max Marks =100

Pass Marks =45

if maths >= 45 and physics >= 45 and chem >= 45:

Exam Result

Challenge



maths



physics



chem

Max Marks =100

Pass Marks =45

if maths >= 45 and physics >= 45 and chem >= 45:

print('Pass')

Exam Result

Challenge



maths



physics



chem

Max Marks =100

Pass Marks =45

```
if maths >= 45 and physics >= 45 and chem >= 45:  
    print( 'Pass' )  
else:
```

Exam Result

Challenge



maths



physics



chem

Max Marks =100

Pass Marks =45

```
if maths >= 45 and physics >= 45 and chem >= 45:  
    print( 'Pass' )  
else:  
    print( 'Failed' )
```

Exam Result

Challenge



maths



physics



chem

Max Marks =100

Pass Marks =45

```
if maths >= 45 and physics >= 45 and chem >= 45:  
    print( 'Pass' )  
else:  
    print( 'Failed' )
```

Exam Result

Challenge



maths



physics



chem

Max Marks =100

Pass Marks =45

```
maths = ?  
physics = ?  
chem = ?  
  
if maths >= 45 and physics >= 45 and chem >= 45:  
    print( 'Pass' )  
else:  
    print( 'Failed' )
```